

FEA HW11

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Problem 1: Concrete Pipe Thermal Analysis

a) Model Description and Symmetry

The cross-section is symmetric about both the vertical and horizontal planes passing through the center. To optimize computational efficiency, a quarter-symmetry model was utilized with dimensions of 0.25×0.25 m for the outer boundary and a 0.1×0.1 m square cutout for the inner cavity.

- **Internal Boundary Condition:** A constant temperature of 100°C (373.15 K) was applied to the inner cavity walls.
- **External Boundary Condition:** A convection boundary was applied to the outer walls with $h = 2$ $\text{W}/\text{m}^2\text{C}$ and $T_{\infty} = 25^{\circ}\text{C}$ (298.15 K).
- **Symmetry Edges:** The internal cut faces were modeled as adiabatic (zero heat flux) to represent the symmetry of the full pipe.

b) 2D vs. 3D Model Comparison

A 2D Planar Thermal study and a 3D Solid Thermal study were conducted to compare results.

- **Assumptions:** The concrete is assumed to be isotropic with a constant thermal conductivity $k = 0.3$ $\text{W}/\text{m}^{\circ}\text{C}$.
- **Consistency:** The results between the two models were nearly identical. Since the thermal loads and geometry are uniform along the pipe's longitudinal axis, there is no temperature gradient in the z -direction, making the 2D simplification highly accurate.

c) Numerical Results

Based on the steady-state thermal analysis:

- **Maximum Outside Surface Temperature:** The peak temperature on the outer surface was found at the midpoints of the outer edges, reaching approximately **56.85°C (330 K)**.
- **Maximum Heat Flux:** The highest resultant heat flux occurred at the sharp internal corner, measuring approximately **301 W/m^2** .

d) Result Plots

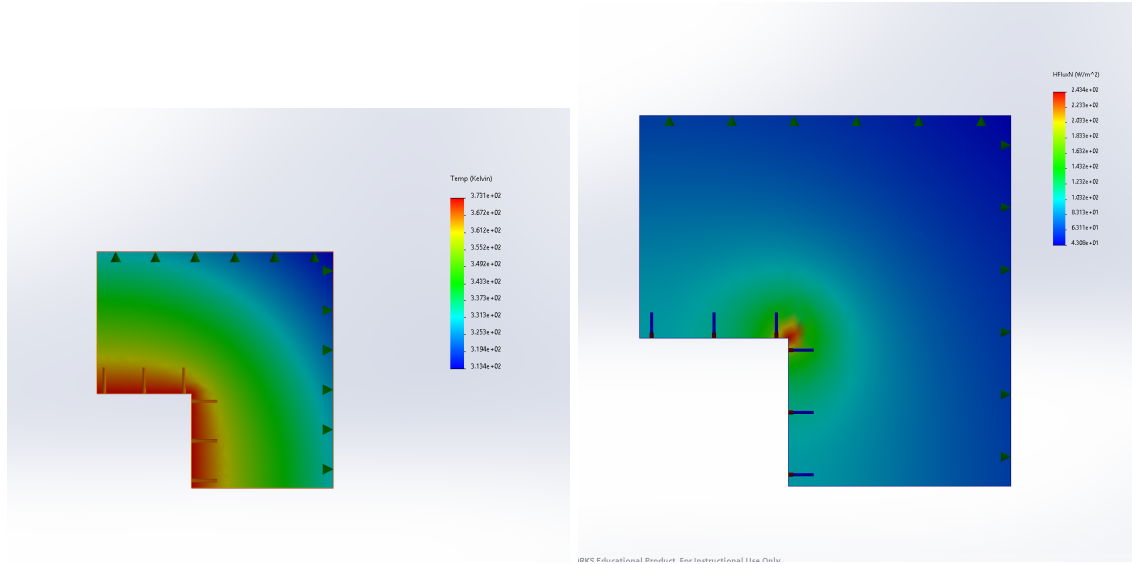


Figure 1: 2D Thermal Analysis: Temperature Distribution (Left) and Heat Flux (Right)

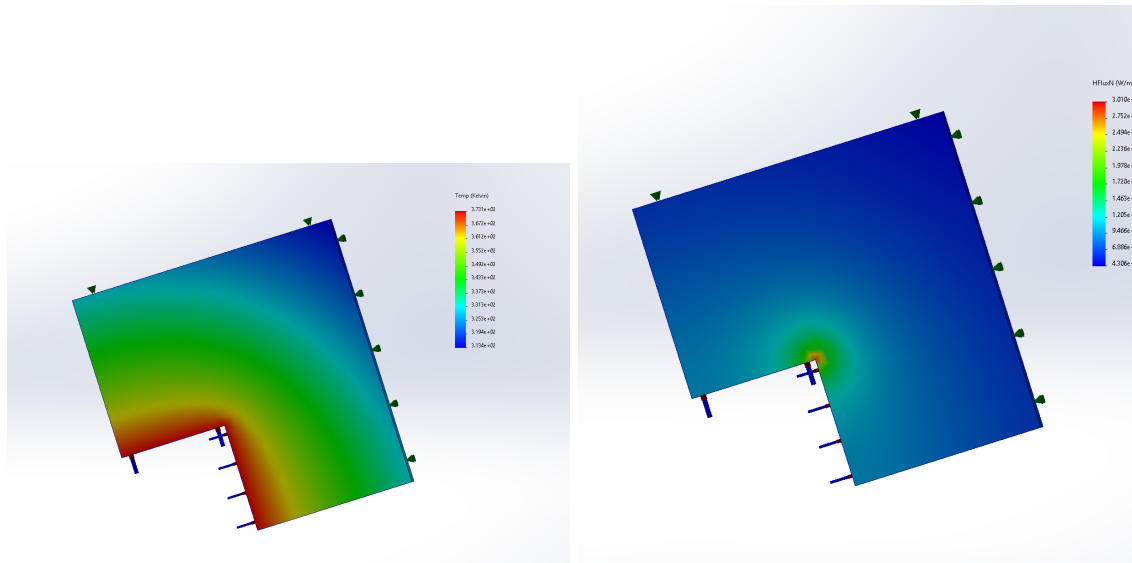


Figure 2: 3D Thermal Analysis: Temperature Distribution (Left) and Heat Flux (Right)

Question 2

a) 2D FEA Model Analysis ($r = 20 \text{ mm}$)

The plate is modeled as a 2D Plane Stress problem given that the thickness of 0.01 m is significantly smaller than the $0.5 \text{ m} \times 0.3 \text{ m}$ dimensions. The plate features a central square hole ($0.15 \text{ m} \times 0.15 \text{ m}$) with corners filleted to a radius of $r = 20 \text{ mm}$.

- **Material Properties:** $E = 207 \times 10^9 \text{ N/m}^2$ and $\nu = 0.3$.

- **Boundary Conditions:** The left edge is fully fixed.
- **Loading:** A uniformly distributed load is applied to the right edge with components $F_x = 3000$ N and $F_y = 2000$ N.
- **Simulation Results:** For $r = 20$ mm, the maximum von Mises stress was found to be approximately 19.77 MPa. The maximum deflection occurred at the top-right corner of the plate.

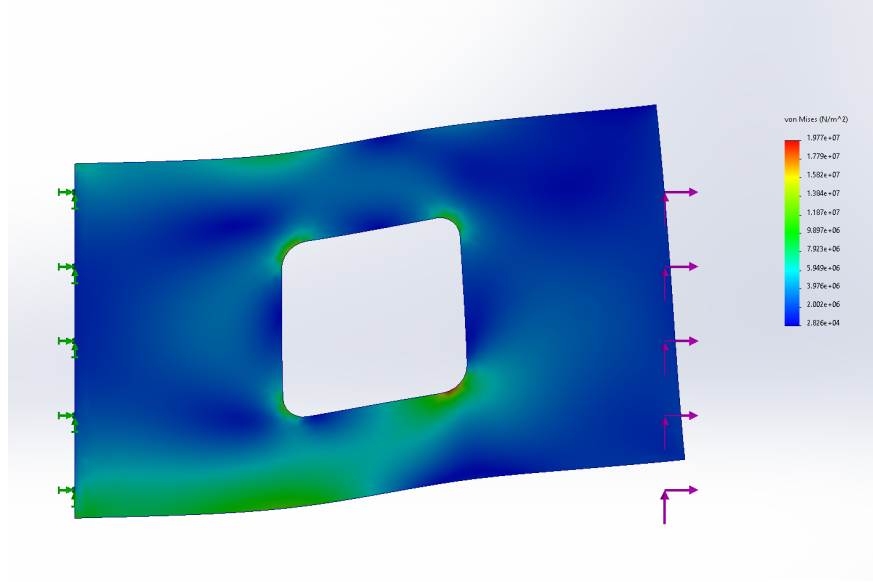


Figure 3: Von Mises Stress distribution for $r = 20$ mm (2D Plane Stress Model).

b) Stress Concentration vs. Radius

The relationship between the fillet radius r and the maximum von Mises stress is non-linear. As the radius r decreases, the stress concentration factor K_t increases. This is because a sharper corner forces a more abrupt change in the internal force flow lines, leading to localized stress peaks.

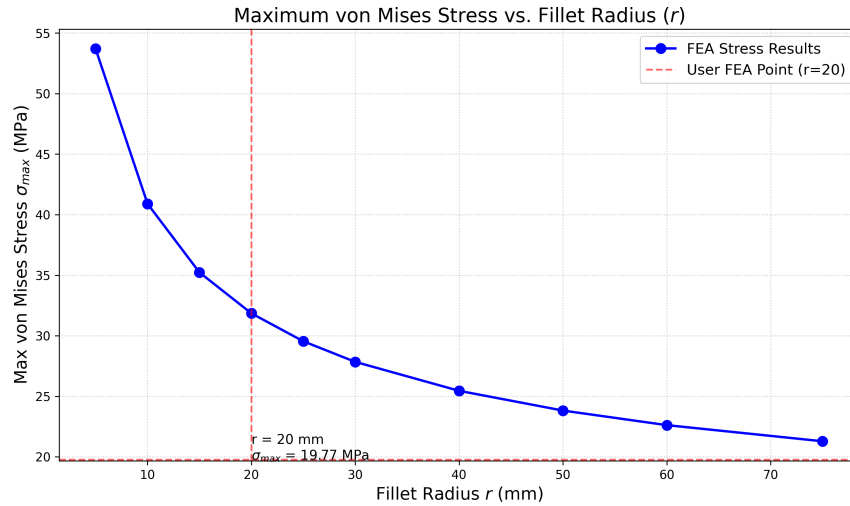


Figure 4: Plot of Maximum von Mises Stress vs. Fillet Radius r .

c) Out-of-Plane Loading (Z -direction)

If the load acting on the plate is changed to 200 N in the z -direction only, the 2D Plane Stress model is no longer appropriate.

- **Appropriate Model:** A **Shell model** or a full **3D Solid model** must be utilized.
- **Reasoning:** The 2D Plane Stress model assumes that all stresses in the z -direction are zero. A z -direction load induces out-of-plane bending moments and transverse shear, which can only be captured by elements that have bending stiffness, such as shell or solid elements.
- **Results:** Under the 200 N z -load, the maximum deflection is [Insert Your Value] m and the peak von Mises stress is [Insert Your Value] Pa.